

STQD-Det: Spatio-Temporal Quantum Diffusion Model for Real-Time Coronary Stenosis Detection in X-Ray Angiography

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Abstract—Detecting coronary stenosis accurately in X-ray angiography (XRA) is important for diagnosing and treating coronary artery disease (CAD). However, challenges arise from factors like breathing and heart motion, poor imaging quality, and the complex vascular structures, making it difficult to identify stenosis fast and precisely. In this study, we proposed a Quantum Diffusion Model with Spatio-Temporal Feature Sharing to Real-time detect Stenosis (STQD-Det). Our framework consists of two modules: Sequential Quantum Noise Boxes module and spatio-temporal feature module. To evaluate the effectiveness of the method, we conducted a 4-fold cross-validation using a dataset consisting of 233 XRA sequences. Our approach achieved the F1 score of 92.39% with a real-time processing speed of 25.08 frames per second. These results outperform 17 state-of-the-art methods. The experimental results show that the proposed method can accomplish the stenosis detection quickly and accurately.

Index Terms—Diffusion model, quantum noise, spatio-temporal feature sharing, stenosis detection, video object detection.

I. INTRODUCTION

CORONARY artery disease (CAD) is one of the deadliest cardiovascular diseases [1], caused by the build-up of atherosclerotic plaque in the coronary artery's inner wall [2]. X-ray angiography (XRA) has a high resolution that facilitates the location of artery stenosis, thus becomes the gold standard for CAD diagnosis in clinical practice [3]. However, doctors usually review XRA images to identify stenosis based on their own experience. The subjective differences among doctors in the CAD judgement impact subsequent treatment decisions [4]. Hence, automated detection of stenosis in XRA images holds

paramount importance in the diagnosis and treatment of CAD. However, automatic identification of coronary artery stenosis rapidly and accurately still faces several challenges: First, due to fluoroscopic projection imaging, the vessels are overlapping in XRA. Furthermore, coronary arteries and lesions move and deform as a result of respiratory and heartbeat movements. These can lead to under-detection of stenosis. Second, the over-detection of stenosis is caused by uneven distribution of contrast medium as well as low contrast between the foreground and background.

Early stenosis detection algorithms traditionally analyzed stenosis based on the results of vessel segmentation [5]. These methods heavily depend on the precision of the segmentation algorithms [6], [7]. Thus, algorithms for directly detecting stenosis [8], [9] have become a hotspot. However, the temporal information provided by XRA sequences is not utilized. These algorithms will yield false detection when the low contrast and overlapping occur in the single XRA frame. Some algorithms [10], [11] try to use transformer or multi-head attention to establish the connections across different frames. These algorithms primarily emphasize local stenosis information and overlook global information. Furthermore, their computational efficiency is relatively low due to the complexity is squared with the input feature.

Recently, diffusion model has demonstrated significant potential in single-image object detection [12]. In this study, we introduce a Quantum Noise Diffusion Model with Spatio-Temporal Feature Sharing for detect Stenosis (STQD-Det). The method is designed to detect stenosis for XRA sequences in real-time. The framework begins with forward diffusion, generating sequential quantum noise boxes (SQNB) and extracting features of images. Following this is the reverse diffusion process. This process consists of three stages. The first stage produces the pre-detecting results by the stenosis detection decoder. Then the spatio-temporal feature sharing (STFS) module is used to share the correct features with the false frames. The third stage is designed to re-detect false frames using the stenosis detector to achieve accurate stenosis detection.

The main contributions in this paper can be summarized as follows: (1) We introduced a diffusion model that incorporates quantum noise and spatio-temporal feature sharing. The model regards the stenosis detection approach as a noise-to-box denoising process, and significantly improves the inference speed and accuracy of the network. To the best of our knowledge,

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it is the first work to integrate quantum noise into diffusion model and apply it to video object detection. (2) We designed a spatio-temporal feature sharing module to share features between correctly detected images and those with errors. This enhancement improves the network's ability to detect stenosis in complex structures and promotes consistency in detection results within the same sequences. (3) We compared STQD-Det with several state-of-the-art object detection methods, including 7 single-frame object detection algorithms, 5 video object detection networks, and 5 vascular stenosis detection methods. Both qualitative and quantitative results demonstrate that STQD-Det can achieve accurate and robust stenosis detection in XRA images, outperforming the compared methods.

II. RELATED WORKS

A. Diffusion Model

As a deep generation model, diffusion model is widely used in computer vision, such as image generation, image segmentation and restoration. Sohl-Dickstein et al. [13] inspired by non-equilibrium statistical physics and proposed the diffusion model, the authors used continuous Gaussian noise and discrete binomial distributions for the validation of the algorithm, proving that both continuous and discrete distributions are feasible in the diffusion model. In image generation, diffusion models [14], [15], [16] execute the forward diffusion process by continuously adding Gaussian noise. The noise can help networks to learn the information-noise distributions of natural images. In image segmentation, Gu et al. [17] were pioneers in proposing an instance segmentation algorithm based on the diffusion model. The segmentation task is regarded as a noise-to-filter denoising process and achieved competitive performance compared to existing instance segmentation models. Rahman et al. [18] leveraged a random sample diffusion model to generate a couple of reasonable segmentation outputs to achieve ambiguous segmentation of multi-modality medical images. Kim et al. [19] introduced a fusion of the diffusion model and adversarial network to generate vessel segmentation masks or pseudo-vascular structures on a pure background image. In image restoration, Wang et al. [20] integrated zero-shot learning into diffusion model to realize super resolution, colorization, inpainting, compressed sensing, and deblurring by solving arbitrary linear IR problem, etc. Hu et al. [21] proposed a fully unsupervised diffusion model with sequential self-fusion, significantly improved the quality of OCT sequence images. Xiang et al. [22] introduced a three-stage self-supervised denoising framework that integrates statistical denoising theory with the diffusion model. This innovative approach significantly enhances denoising performance and improves the quality of MRI images with low signal-to-noise ratio (SNR). In the field of medical image processing, the anomaly detection is an important task. Health images are generated from lesion images by using diffusion model. The lesion area is further obtained by subtraction with the lesion image [23], [24], [25], [26]. This method heavily relies on the reliability of the generated healthy images. Chen et al. [12] initially employed the diffusion model for object detection, conceptualizing the object detection task as a noise-to-box transformation. The detection

boxes in this work have non-continuous coordinates and scales. However, it still employed continuous Gaussian distribution to generate noise boxes. Recently, some researchers have tried to introduce different noises into the diffusion model and verify the performance. Nachmani et al. [27] introduced gamma noise into the diffusion model, the generative performance of the diffusion model is improved by mixing the sampling of gamma noise with Gaussian noise. Bansal et al. [28] replaces the noise in the diffusion model with noiseless/cold image transforms and achieved good image generation results. In Diffusion Schrödinger Bridge (DSB) model [29], [30], which treats the diffusion process of a diffusion model as a Schrödinger bridge problem. Robust generation is achieved by finding the optimal interpolation path between the two distributions through a stepwise deterministic diffusion process.

B. Stenosis Detection in XRA Images

The abnormal change in vascular radius is the most obvious sign of stenosis in XRA images. To this end, Wan et al. [5] proposed a comprehensive framework to detect vascular stenosis. This approach includes several key steps, including image enhancement using a Hessian-based algorithm, vessel skeleton extraction using the level set algorithm, estimation of vessel radius and direction through model matching, and identification of stenosis locations via local extremum detection. The multi-stage framework accumulates errors at each step and reduces detection accuracy. To avoid the above problem, Ovalle et al. [9] integrated quantum network with the classical convolutional neural network (CNN) ResNet. This hybrid approach enhanced CNN's performance. Nevertheless, this algorithm primarily relied on single-frame images and performed poorly when frames exhibited issues such as overlapping, high noise levels, or limited foreground-background distinction. Addressing these challenges, algorithms needed to leverage temporal context provided by XRA sequences. Wu et al. [8] proposed a sequence-false positive suppression module, in which utilized the temporal consistency of consecutive frames to suppress false positive detection in partial XRA frames. Pang et al. [10] introduced an attention module designed to fuse features from sequence frames and correlate temporal context information. Additionally, a sequence consistency alignment module was employed to suppress false positive detection and supplement false negative detection. However, the sequence consistency alignment module has high sensitivity to vessel motion amplitude. Han et al. [11] were pioneers in integrating the Transformer into coronary stenosis detection, introducing a novel spatio-temporal aggregation framework known as Spatio-Temporal Feature Aggregation. They achieved this by capturing spatio-temporal region of interest (RoI) features within a localized window, generating corresponding tokens, and employing the Transformer to aggregate these spatio-temporal features, ultimately yielding accurate stenosis detection results. However, the computational demands of the Transformer and the adjustment speed of anchors through the Regional Proposal Network (RPN) resulted in slow inference speeds, rendering it unsuitable for real-time stenosis lesion detection.

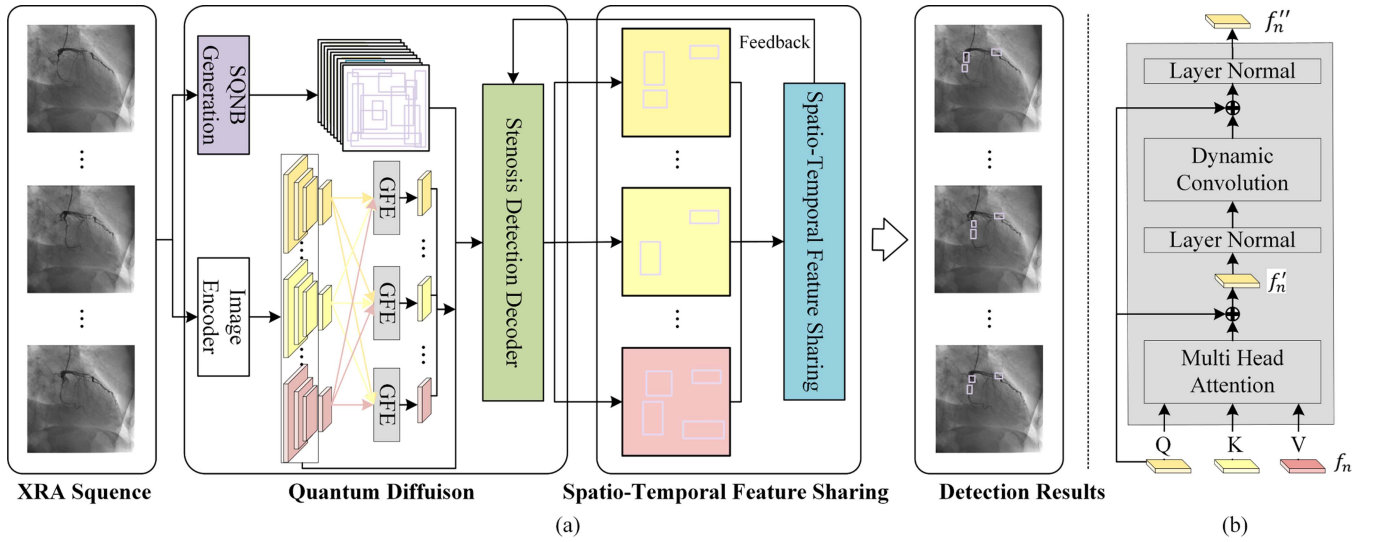


Fig. 1. Overview of STQD-Det. (a) is the flow chart of proposed method, (b) is the structure of Global Feature Enhancement (GFE) block.

III. METHODS

A. Overview

The overall STQD-Det architecture is shown in Fig. 1. It contains two main components: Quantum Diffusion (QD) model and Spatio-Temporal Feature Sharing (STFS) module. The QD model consists of forward diffusion and reverse diffusion process. In the forward diffusion process, the Sequential Quantum Noise Box (SQNB) Generator is used to form noise boxes. These noise boxes can be regarded as candidates for the stenosis, which represent the stenosis regions of interest for the subsequent denoising process. At the same time, the Image Encoder with Global Feature Enhancement (GFE) is for extracting features from input images and then aggregate the highest layer of feature maps. In the reverse diffusion process, the Stenosis Decoder is used for the noise-to-box process and outputs the primary detection results. The primary results are then transmitted to the STFS module. In STFS, the features from positive frames are shared with the negative ones in the same sequence. Here, the frames with over- and under-detection of lesions can be considered as negative frames, and the remaining frames are considered as positive frames. After that, these enhanced feature maps are fed back into the Stenosis Decoder to produce refined results.

B. Quantum Diffusion Model

The primary noise prevalent in X-ray images is quantum noise, which exhibits a discrete distribution due to the discrete energy of quantum photons. Similarly, the coordinates of noise boxes also follow a discrete-type distribution. To this end, we have introduced the quantum noise into diffusion model for coronary stenosis detection. The diffusion processes of the quantum noise diffusion model are shown in Fig. 2, it consists of two process. Diffusion model consists of two processes, the forward process and the reverse diffusion process. The forward diffusion

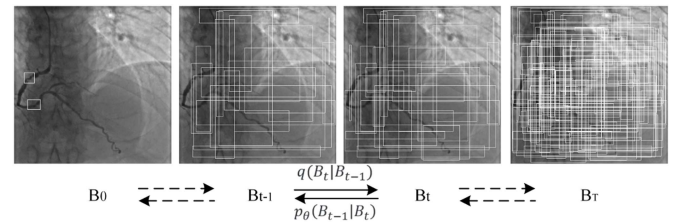


Fig. 2. Diffusion process of the sequential quantum noise diffusion model.

process transforms the original noise boxes' distribution B_0 into the final noise boxes' distribution B_T through step-by-step iteration, and the reverse process transforms the B_T back to the B_0 . Markov chains can be used to model the forward process, which is designed to transit from $q_0(B_0)$ to the distribution that approximates $q_T(B_T)$ in T intermediate steps. In the chains, $q_{t|t-1}(B_t|B_{t-1})$ denotes the transformation of the distribution from the $t-1$ step to the t step. B_t is the distribution of the noise boxes in t th time step, $t \in [0, T]$, T is the maximum step of the chain. Since quantum noise obeys the Poisson distribution $\mathcal{P}(\lambda)$, the probability density of forward diffusion process can be formally defined as:

$$q(B_{0:T}) = B_0 \prod_{t=0}^T q_{t|t-1}(B_t|B_{t-1}) \quad (1)$$

where:

$$\begin{aligned} q_{t|t-1}(B_t|B_{t-1}) &= \mathcal{P}(B_t; \lambda_t B_{t-1}) \\ &= \frac{\lambda_t^{B_t} e^{-\lambda_t}}{B_t!} \end{aligned} \quad (2)$$

where $\lambda_t = B_{t-1} + \gamma_t f_t(B_t)$ is the parameter of the Poisson distribution at step t , γ_t is the factor to control the scale of transition and $f_t(B_t)$ is the function that describes the trend of the noise at step t . After step-by-step iteration, the noise

distribution in the final time step is:

$$\begin{aligned}
 B_T &= q(B_T|B_{T-1}) B_{T-1} \\
 &= q(B_T|B_{T-1}) q(B_{T-1}|B_{T-2}) B_{T-2} = \dots \\
 &= q(B_T|B_{T-1}) q(B_{T-1}|B_{T-2}) \dots q(B_1|B_0) B_0 \\
 &= \frac{\lambda_T^{B_T} e^{-\lambda_T}}{B_T!} \frac{\lambda_{T-1}^{B_{T-1}} e^{-\lambda_{T-1}}}{B_{T-1}!} \dots \frac{\lambda_1^{B_1} e^{-\lambda_1}}{B_1!} B_0
 \end{aligned} \quad (3)$$

where q is the forward diffusion process.

In coronary artery stenosis detection, there is minimal variation in vessel motion amplitude and stenosis position between consecutive frames. In an N -frames XRA sequence, quantum noise boxes are randomly generated and introduced only in the first frame. For the subsequent $N - 1$ frames, the noise distribution can be obtained through perturbation according to the noise distribution of the first frame. Such process can be represented as:

$$\begin{aligned}
 q(B_{n,t}|B_{n,0}) &= \begin{cases} \prod_{t=1}^T q(B_{1,t}|B_{1,t-1}) B_{1,0}, & \text{if } n = 1 \\ \frac{q(B_{1,t}|B_{1,0})}{B_{1,0}} \times B_{n,0}, & \text{if } n = 2, \dots, N \end{cases}
 \end{aligned} \quad (4)$$

where $B_{n,t}$ is the noise distribution of n th frame in t th time step. Inspired by DiffusionDet [12], in the training phase, we generate the noise boxes around the ground truth; While in the inference phase, the noise boxes are around the center of this image. Then, the sequential quantum noise boxes (SQNB) are generated.

Meanwhile, the input sequence images are fed into the image encoder. It uses ResNet-50 as the backbone and Feature Pyramid Network (FPN) to extract multiscale feature maps. With the increase in layers, the features also transition from detailed information to low-level information. The low-level information only contains the position of vessels. To catch the changes in the location of coronary artery, Global Feature Enhancement (GFE) is performed to the highest layer feature maps $F = \{f_n\}_{n=1}^N$, where f_n is the highest feature map of the n th frame. The GFE contains Multi-Head Attention (MHA) block and Dynamic Convolution (DC) [31] block. Compared with ordinary convolutional blocks, the advantage of dynamic convolution lies in its ability to dynamically adjust attention and adaptively modify the weights of the convolutional kernel according to the input data. This capability enhances the model's feature extraction for various inputs. In the stenosis detection task, lesion specificity results in varying lesion sizes, locations, and stenosis grades. In order to improve the feature extraction ability of the model for different lesions, we adopt dynamic convolution block instead of ordinary convolution in the GFE module. The input feature maps are first tokenized as the feature vectors v_n , and then aggregate all the feature vectors into a group $\mathcal{V} = \{v_n\}_{n=1}^N$,

$$\nu_n = \begin{cases} [v_n, v_n, v_{n+1}], & n = 1 \\ [v_{n-1}, v_n, v_{n+1}], & 2 \leq n < N \\ [v_{n-1}, v_n, v_n], & n = N \end{cases} \quad (5)$$

Then the vector group is fed into MHA block. It consists of MHA mechanism, Layer Normalization (LN) and Residual Connection to obtain enhanced features. It can be represented as:

$$v'_n = \text{LN}(\text{MHA}(\nu_n) + v_n), \quad n = 1, 2, \dots, N \quad (6)$$

In MHA block, the three elements in ν_n are used as key, query, and value, respectively. Then the vectors with attention weight are reshaped into attention feature maps $F' = \{f'_n\}_{n=1}^N$, in which f'_n is attention feature map of the n th image. Then they are fed into DC block. This block consists of DC layer, LN layer, Full Connected (FC) layer, and use residual connection to obtain the final feature:

$$f''_n = \text{LN}(\text{FC}(\text{DC}(f_n, f'_n)) + f_n), \quad n = 1, 2, \dots, N \quad (7)$$

where f''_n is GFE feature map of the n th frame. It is merged with the other three layers and performs the first stage reverse process together.

The reverse diffusion process can also be modeled by the Markov chains in the opposite direction, and the transformation of the distribution from the $t - 1$ step to the t step can be denoted as $p_{t-1|t}(B_{t-1}|B_t)$. Thus, the overall reverse process can be expressed as:

$$p(B_{T:0}) = p_T(B_T) \prod_{t=1}^T p_{t-1|t}(B_{t-1}|B_t) \quad (8)$$

where:

$$p_{t-1|t}(B_{t-1}|B_t) = \frac{p_{t|t-1}(B_t|B_{t-1})p_{t-1}(B_{t-1})}{p_t(B_t)} \quad (9)$$

By converting the fractional equation into an exponential term to simplify the (9) and bringing the Poisson distribution into the equation, the expression for the reverse process can be obtained:

$$\begin{aligned}
 p_{t-1|t}(B_{t-1}|B_t) &= p_{t|t-1}(B_t|B_{t-1}) \exp(\log p_{t-1}(B_{t-1}) - \log p_t(B_t)) \\
 &= \frac{1}{B_t!} (B_{t-1} + \gamma_t f_{t-1}(B_{t-1}))^{B_t} e^{-(B_{t-1} + \gamma_t f_{t-1}(B_{t-1}))} \\
 &\quad \times \exp\left(\log\left(\frac{\lambda_{t-1}^{B_{t-1}} e^{-\lambda_{t-1}}}{B_{t-1}!}\right) - \log\left(\frac{\lambda_t^{B_t} e^{-\lambda_t}}{B_t!}\right)\right)
 \end{aligned} \quad (10)$$

Let's further simplify the logarithmic part, which yields:

$$\begin{aligned}
 p_{t-1|t}(B_{t-1}|B_t) &= \frac{1}{B_t!} (B_{t-1} + \gamma_t f_{t-1}(B_{t-1}))^{B_t} e^{-(B_{t-1} + \gamma_t f_{t-1}(B_{t-1}))} \\
 &\quad \times \exp((B_{t-1} \log \lambda_{t-1} - \lambda_{t-1} - \log(B_{t-1}!)) \\
 &\quad - (B_t \log \lambda_t - \lambda_t - \log(B_t!))) \\
 &= \frac{1}{B_t!} (B_{t-1} + \gamma_t f_{t-1}(B_{t-1}))^{B_t} e^{-(B_{t-1} + \gamma_t f_{t-1}(B_{t-1}))} \\
 &\quad \times \exp(B_{t-1} \log \lambda_{t-1} - \lambda_{t-1} - B_t \log \lambda_t - \lambda_t)
 \end{aligned} \quad (11)$$

From this, we introduce specific expressions for the forward and reverse processes when deploying Quantum noise into the diffusion model. In this case, the reverse process is accomplished

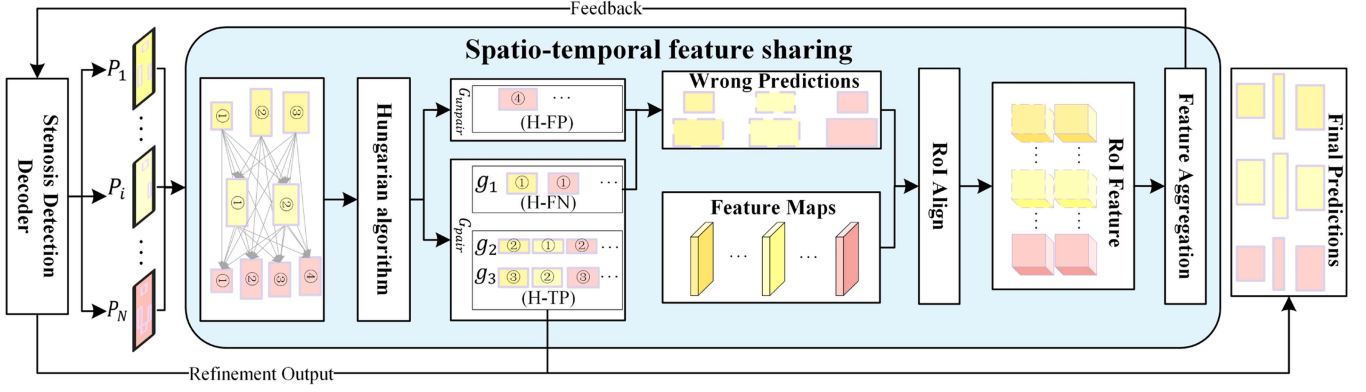


Fig. 3. The flow chart of spatio-temporal feature sharing module.

by the deep learning model through learning the forward process. The deep learning network fits the noise distribution derived from (11) by tuning its own parameters θ , then the distribution of the quantum noise boxes in the reverse process can be expressed as p_θ . Thus, the Markov chain of the reverse process can be expressed in a learnable form:

$$p_\theta(B_{t-1}|B_t) = \mathcal{P}(B_{t-1}; \lambda_t B_t) \quad (12)$$

If $\mathcal{P}(B_T)$ of final time step is known, the network can infer the reverse process of restoring the stenosis detection boxes from the noise boxes by combining the feature maps. The total time steps reverse process can be expressed as:

$$p_\theta(B_{0:T}) = \prod_{t=1}^T p_\theta(B_{t-1}|B_t) \quad (13)$$

Each time step in the Markov chain represents the process of the network learning to recover the stenosis boxes from the noise boxes gradually. The quantum noise box and the corresponding global enhancement feature for each image frame serve as inputs to a stenosis detection decoder. This decoder undergoes learning in the reverse diffusion process. The structure of the decoder comes from Diffusion Det [12], it takes the feature map and SQNB as input, first extracts the RoI feature corresponding to the detection box from the feature map through RoI Align [32]. Subsequently, it predicts the category and position of the candidate box through the classification and regression heads, ultimately achieving the reversal of the diffusion process.

C. Spatio-Temporal Feature Sharing

The presence of stenosis within a given XRA sequence should exhibit consistency. However, due to the motion of coronary arteries or insufficient contrast medium, missed detection or over-detection may occur in certain frames. The Spatio-Temporal Feature Sharing (STFS) module is proposed to ensure the consistency of outputs by sharing the spatio-temporal RoI features in positive frames to the negative frames.

The proposed STFS module is shown in Fig. 3. First, we extract the box $P_n^{s_n}$, which means the s th box in the n th frame in $\{P_1, P_2, \dots, P_n, \dots, P_N\}$. Then constitute them into a candidate pool $\{P_n^{s_n}\}_{n=1}^N (s_n = 1, 2, \dots, S_n)$, in which S_n

is the number of candidates in current frame. Second, the frame-to-frame matching of candidates is performed by using the Hungarian algorithm [33], the matching function can be defined as:

$$m_f(P_i^{s_i}, P_j^{s_j}) = (o_{i,j}, d_{i,j}), \quad i \neq j \in 1, 2, \dots, N \quad (14)$$

where $d_{i,j}$ is the Manhattan Distance between $P_i^{s_i}$ and $P_j^{s_j}$, $o_{i,j}$ reflects the overlap rate of the two boxes. The overlap rate, calculated by Intersection over Union (IoU) [30] within the value range of [0, 1], indicates a higher degree of overlap with a higher value. Then the Hungarian algorithm uses $o_{i,j}$ as the primary weight and $d_{i,j}$ as the secondary weight to divide the candidate pool into the paired group and unpaired group. In cases where there are candidate boxes remaining in the unpaired group, they will be matched with the candidate boxes in the next neighboring frame. This process continues until deplete all the frames. At this point, the paired group is defined as $G_{pair} = \{(P_i^{s_i}, P_j^{s_j}), (P_j^{s_j}, P_k^{s_k}), \dots\}_{i \neq j \neq k \in 1, 2, \dots, N}$, while the unpaired group consists of all candidate frames that were not successfully paired.

To obtain the stenosis information of the whole sequence, we combine the candidate boxes in the paired group by element splicing:

$$\text{concat}((P_i^{s_i}, P_j^{s_j}), (P_j^{s_j}, P_k^{s_k})) = (P_i^{s_i}, P_j^{s_j}, P_k^{s_k}) \quad (15)$$

Element within the paired group will contain all the candidate box sequences that can be grouped into a box: $G_{pair} = \{g_1 = (P_1^{s_1}, \dots, P_n^{s_n}, \dots, P_N^{s_N}), g_2, \dots\}$. After that, count the number of candidate boxes n_{boxes} contained in each paired group, and divide these sequences again by voting: If $n_{boxes} = N$, indicating that all images contain this detection box, the corresponding sequence is stored in the Hypothesis true positive (H-TP) group; If $\frac{N}{2} \leq n_{boxes} < N$, it means that the position is detected in most images, only some images do not contain this detection box, and these sequences are stored in the Hypothesis false negative (H-FN) group; If $n_{boxes} < \frac{N}{2}$, indicating that only a few images contain this detection box, these sequences are stored in the Hypothesis false positive (H-FP) group together with the detection boxes in the unpaired group. The detection boxes in the H-TP group will be used directly for network output. On the other hand, the detection boxes in the unpaired group will

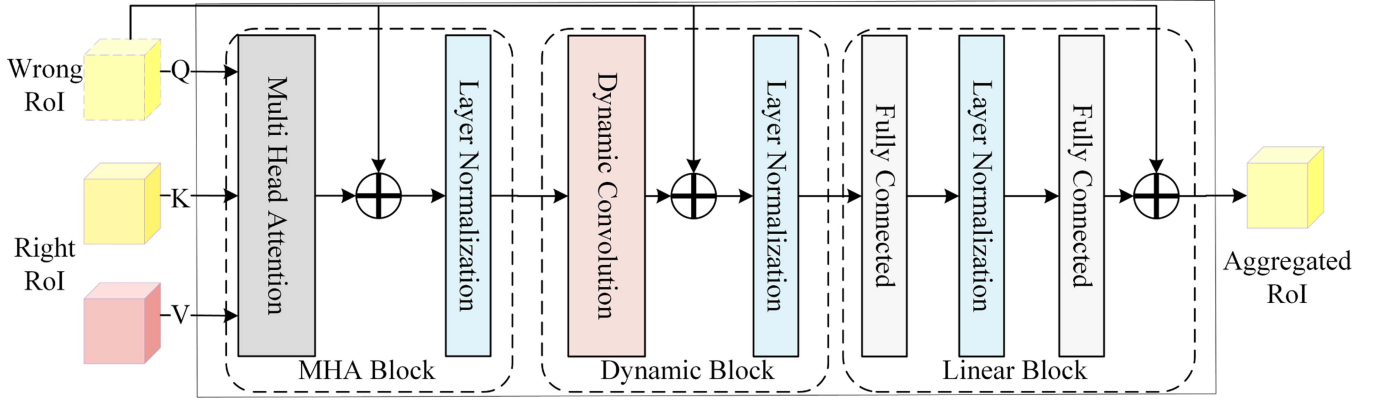


Fig. 4. Structure diagram of RoI feature aggregation module.

be treated as the second-stage noise boxes and sent to the RoI feature aggregation section. The images containing missing or redundant candidate boxes are considered as error frames, while the remaining images are considered as reference frames.

During aggregation, the RoI features correspond to the positive frames are projected onto the negative frames. RoI padding is performed first to avoid the effects caused by vessel motion. The box can be expressed as $P_n^{s_n}(x_{1,n}, y_{1,n}, x_{2,n}, y_{2,n})$, where $x_{1,n}$ and $y_{1,n}$ denote the abscissa and ordinate of the upper left corner of $P_n^{s_n}$, and $x_{2,n}$ and $y_{2,n}$ denote the lower right corner of $P_n^{s_n}$. Then the RoI feature of H-FN group can be defined as $P_{FN}^{s_n}(x_{1,FN}, y_{1,FN}, x_{2,FN}, y_{2,FN})$, where:

$$x_{1,FN}, y_{1,FN} = \min_{n=1,2,\dots} x_{1,n}, \min_{n=1,2,\dots} y_{1,n} \quad (16)$$

$$x_{2,FN}, y_{2,FN} = \max_{n=1,2,\dots} x_{2,n}, \max_{n=1,2,\dots} y_{2,n} \quad (17)$$

Considering that there may be only one candidate in H-FP, the RoIs are defined as $P_{FP}^{s_n}(x_{c,FP}, y_{c,FP}, w_{FP}, h_{FP})$, where $x_{c,FP}$ and $y_{c,FP}$ represent the abscissa and ordinate of the center point of P_{FP}^s , respectively, and w_{FP} and h_{FP} represent the width and height of P_{FP}^s , respectively:

$$x_{c,FP}, y_{c,FP} = \frac{\sum_{j=1}^n x_{2,j} + x_{1,j}}{2n}, \frac{\sum_{j=1}^n y_{2,j} + y_{1,j}}{2n} \quad (18)$$

$$w_{FP}, h_{FP} = \alpha \frac{\sum_{j=1}^n x_{2,j} - x_{1,j}}{n}, \alpha \frac{\sum_{j=1}^n y_{2,j} - y_{1,j}}{n} \quad (19)$$

α is the padding coefficient, which is used to control the size of the RoI region. In this paper, $\alpha = 2$ is taken in combination with the range of vascular motion. Then use RoI Align to cut out the corresponding RoI feature and feed into the RoI aggregator. The aggregator is composed of MHA block, DC block and Linear block, as shown in Fig. 4. The MHA block and DC block are as same as the GFE, while the linear block can be described as:

$$f_n''' = FC_2(ReLU(FC_1(f_n''))) + f_n \quad (20)$$

where f_n is the origin RoI feature of the wrong frame, f_n'' is the output of DC block, and f_n''' is the final output of RoI aggregator. FC_1 and FC_2 are the fully connected layer and use Rectified

Linear Unit (ReLU) as the activation function. Finally, feed the aggregated RoI feature into stenosis decoder again to accomplish the second stage reverse diffusion and output accurate detection results.

To strengthen the network's capacity for maintaining consistency in the count of stenosis detection boxes, during the training stage, the network will reevaluate the count of detection boxes, denoted as $n_{boxes,n}$, for each image in the sequence. A consistency loss is then employed to enforce this constraint, and is defined as follows:

$$L_{num} = \frac{\sum_{n=1}^N |n_{boxes,n} - n_r|}{N} + \beta \quad (21)$$

where n_r is the number of detection boxes per image determined by voting in the above process, and β is a constant to prevent loss from being zero. The STFS module plays a crucial role in enhancing the network's capacity to differentiate between weak lesions and non-lesions, ultimately improving the accuracy of stenosis detection.

IV. EXPERIMENTAL RESULTS

A. Datasets

In the experiments, we collected 233 XRA sequences from Peking Union Medical College Hospital in China. The frame rate is 15 FPS, the size is 512×512 and the resolution is $0.3 \times 0.3 \text{ mm}^2$. During the experiments, we extracted 9 consecutive contrast-filled frames as input. The stenosis is labeled by three experienced interventional radiologists and then integrated them as the ground truth. To verify the stability and accuracy of the framework, a four-fold cross-validation is used to divide the data set randomly, and the ratio of training set to test set is 3:1.

B. Evaluation Criteria

To evaluate the performance of the proposed method, we used Precision ($Prec$), Sensitivity ($Sens$), F_1 score (F_1) and Frame Per Second (FPS). The four metrics are defined as follows:

$$Prec = \frac{TP}{TP + FP} \quad (22)$$

$$Sens = \frac{TP}{TP + FN} \quad (23)$$

$$F_1 = 2 \times \frac{Prec \times Sens}{Prec + Sens} \quad (24)$$

$$FPS = \frac{N_{images}}{T_{total}} \quad (25)$$

where N_{images} is the total number of inferred images, T_{total} is the total time that consumed by inference. We also used Hungarian algorithm to match the prediction boxes with ground truth boxes and set the threshold of IoU as 0.5, which means match failed when the IoU between prediction boxes and ground truth boxes is less than 0.5.

C. Implementation Detail

In the proposed framework, the backbone ResNet50 is initialized with the pre-trained weights on ImageNet-1K [34] and the other parts of the framework is initialized by Xavier. To optimize the weights of the proposed method, the Adaptive Moment Estimation with decoupled weight decay (AdamW) optimizer is adopted in the training phase, which has a learning rate of 2.5×10^{-5} and a weight decay of 1×10^{-4} . For convergence, a linear warm-up is applied in the first 500 iterations in the training and the total iteration number is 25000. The batch size of this framework is set to 1 and each mini-batch contains 9 consecutive contrast-filled frames in the same XRA sequence. The proposed framework is running on Ubuntu 20.02 and uses a single NVIDIA GeForce RTX 4090 GPU to train the network.

In the inference phase, we use stenosis decoder to predict the location and scale of the quantum noise boxes iteratively. The final prediction result is obtained after Non-Maximum Suppression (NMS). To ensure that the FP detection are suppressed while retaining the TP detection frames as much as possible, a threshold of 0.5 is used for the confidence score. In other words, detection frames with a score lower than 0.5 will be suppressed, while those with a score of 0.5 or higher will be retained.

D. Ablation Experiment

To assess the significance of the QD and STFS module within STQD-Det, we conducted ablation experiments to analyze their impact on the detection performance of coronary stenosis lesions. By removing these two modules, we obtained the basic framework (Baseline) of the proposed method. In the Baseline, the traditional Non-sequential Gaussian Noise Box (NSGNB) is utilized instead of SQNB to initialize the stenosis candidate box. To verify the enhancement of model performance by quantum noise compared to Gaussian noise in the diffusion-based object detection model, non-sequential quantum noise boxes (NSQNB) are used instead of NSGNB on the basis of Baseline method, which can be denoted by Baseline + NSQNB. On the basis of Baseline, sequential Gaussian noise is used instead of the original NSGNB to verify the improvement of the sequential noise detection boxes on the model detection performance in the VOD tasks. And it can be represented by Baseline + SGNB. The elimination of STFS is denoted as W/o STFS, and the removal of QD is represented by Baseline + STFS.

TABLE I
COMPARISON OF THE NETWORK VARIANTS FOR STENOSIS DETECTION ON THE XRA IMAGES OF CORONARY ARTERY

Methods	Prec	Sens	F_1
Baseline	86.89	74.36	82.86
Baseline + NSQNB	91.17	78.48	84.35
Baseline + SGNB	86.49	80.05	83.15
W/o STFS	91.20	80.39	85.48
Baseline + STFS	93.52	82.05	87.28
STQD-Det	94.52	90.46	92.45

The quantitative results of stenosis detection in XRA images by the proposed STQD-Det and three ablation methods are shown in Table I. STQD-Det achieves the best performance in *Prec*, *Sens* and F_1 , which are 90.46%, 94.52%, and 92.45%, respectively. While the baseline performs poorly on these three metrics due to its inability to leverage spatio-temporal information within the sequence. The Baseline + NSQNB method shows a greater improvement of 4.28% in *Prec* compared to the Baseline method, which is due to the fact that the noise detection box generated by discrete quantum noise is more broadly distributed in terms of scale and location, allowing the network to capture more features of stenotic lesions that are located in different locations and at different scales, thus producing fewer FP detections. The Baseline + SGNB method, on the other hand, has improved 5.69% in *Sens* because the noise detection boxes with sequence prior enables the model to focus on the same locations in the same video sequence, which aids the model in understanding the motion of stenotic lesions in different frames and reduces the number of FN detections. The W/o STFS method can be seen as a combined version of the two previous methods. Compared to the baseline method, there is an improvement in all three metrics. However, it still exhibits inferior performance compared to the proposed method. Particularly, it underperforms in *Sens* and F_1 , which are 10.07% and 6.97% lower than proposed method. This is primarily because W/o STFS does not aggregate the local features of stenosis, resulting in the inability to accurately locate weak stenosis caused by vessel motion. This underscores the importance of STFS module in enhancing stenosis detection performance. Baseline + STFS method aggregates the local features of stenosis, but the prediction accuracy is still lower than the proposed method. This is because the randomly initialized detection boxes do not contain sequence prior, which makes the network pay attention to different parts in different frames. Furthermore, the size and location of the detection boxes produced by a continuous distribution exhibit very little change. This makes it challenging for the network to capture features of stenosis lesions with varying scales and positions. Weak stenosis can only be detected in a few frames, which finally results in FN detection. The validity of the QD module is verified. The proposed method not only considers the prior of the sequence images, but also provides the spatio-temporal features from the correct frames for the images with errors, so that the network can better understand the global and local spatio-temporal context information.

Fig. 5 illustrates the qualitative results of the proposed method along with the outcomes of ablation experiments. In the figure,

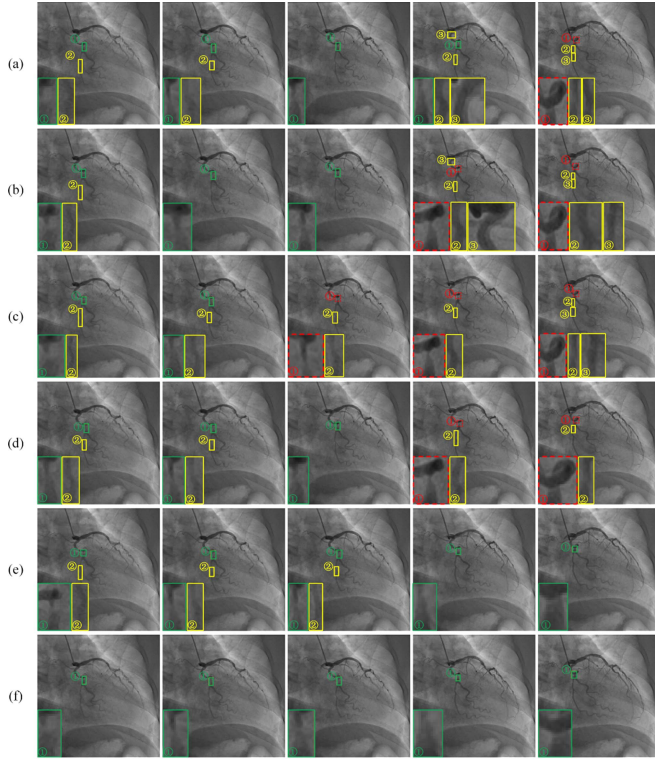


Fig. 5. Qualitative comparison of the proposed method with ablation experiments. (a)–(f) indicate the stenosis detection results of Baseline, Baseline + NSQNB, Baseline + SGNB, W/o STFS, Baseline + STFS and STQD-Det, respectively.

TP, FP, and FN detection boxes are denoted by green solid boxes, yellow solid boxes, and red dashed boxes, respectively. In sequence (a), the Baseline performs poorly due to lack sequence’s prior knowledge and fail to refine the prediction result using information from other frames after generating false detection. In the fifth frame, lesion ① is obscured by other vessels due to vessel motion. It’s a high likelihood of FN detection without referencing the temporal context from other frames. Both the Baseline, Baseline + NSQNB, Baseline + SGNB, and the W/o STFS fail to detect this lesion. In contrast, the proposed method and Baseline + STFS successfully detect it due to the local feature enhancement of STFS. In sequence (b), the Baseline + NSQNB method is able to better focus on the lesion scale and location information, and predicts the lesion ① in the first three images. While in sequence (c), the Baseline + SGNB method performs better in terms of detection consistency, predicting lesion ② on all five images (even though it is wrongly detected), which verifies that the method is able to acquire certain sequence a priori knowledge in the video sequence detection task. In sequence (e), some frames yield FP detections. These are pseudo-stenosis lesions arising from vessel bending and incomplete contrast agent filling. Baseline + STFS exhibits varying detection results in each frame image due to the absence of prior information provided by the QD. The experimental results affirm that the QD and STFS module effectively leverage spatiotemporal context information from XRA sequences, resolving false detections from vascular

TABLE II
COMPARISON OF SINGLE-IMAGE OBJECT DETECTION METHODS

Methods	<i>Prec</i>	<i>Sens</i>	F_1	<i>FPS</i>
RetinaNet	78.25	69.74	73.75	8.44
Faster RCNN	87.36	75.11	80.77	8.72
Cascade R-CNN	89.74	79.40	84.25	7.53
FoveaBox	82.09	80.62	81.62	9.36
DETR	16.83	10.25	12.73	7.97
Deformable DETR	77.87	85.48	81.49	10.24
DINO	86.84	85.71	86.27	16.26
STQD-Det	94.52	90.46	92.45	25.08

motion and incomplete contrast agent filling. This leads to a notable enhancement in the stenosis detection performance of the proposed method.

E. Comparison With Single-Image Object Detection Network

To evaluate the effectiveness of the proposed method for exploiting sequence information, we compare it with seven single-image object detection methods, which contain RetinaNet [35], Faster R-CNN [36], Cascade R-CNN [37], FoveaBox [38], DETR [39], Deformable DETR [40] and DINO [41]. Table II displays the quantitative results of the mentioned methods, with STQD-Det outperforming the others in all metrics. Since the compared methods are primarily tailored for single-image detection tasks, they are not optimized for speed, leading to relatively slow detection processes. DETR performs poorly in almost all metric, primarily due to its approach of achieving spatio context by calculating attention over full image. This approach doesn’t converge effectively on the small dataset, making it challenging to model the characteristics of stenosis. And the other comparison methods lack the temporal information provided by XRA sequence. RetinaNet, Faster R-CNN, Cascade R-CNN, and FoveaBox are based on CNN, which are limited to acquiring local spatial information and result in reduced accuracy. The remaining three methods employ Transformer encoder-decoder architecture, which lack local inductive bias and struggle to model the intricate features of stenosis, leading to lower performance metrics. The proposed method leverages the temporal information offered by XRA images through the introduction of GFE and enhances its comprehension through the STFS module. This results in an improvement in stenosis detection.

The qualitative results are shown in Fig. 6. The RetinaNet method and the FoveaBox method shown in Fig. 6(a) and (d) are single-stage convolutional object detection methods. Such methods sacrifice part of the detection and feature extraction performance to improve the detection speed, thus producing more false detections and missed detections. Among them, RetinaNet is only sensitive to severe stenotic lesions, such as lesion ④, and cannot detect other mild lesions. The Faster RCNN and Cascade RCNN methods shown in Fig. 6(b)–(c) are two-stage Anchor-based Target Detection Networks, which are slower but have improved detection accuracy. Such methods are based on manually designed anchor boxes for target detection, and they are limited by the size of the anchor boxes,

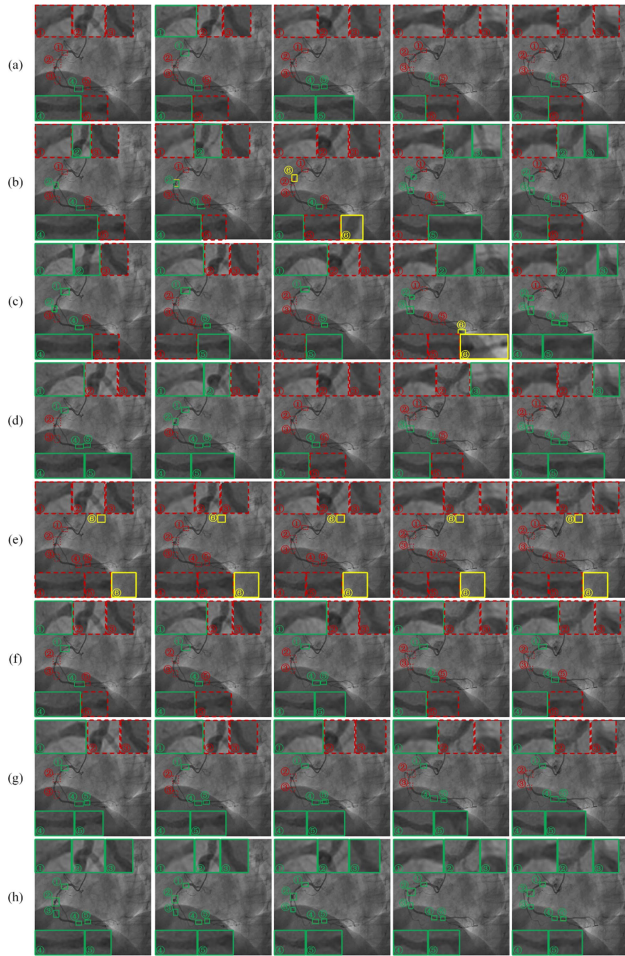


Fig. 6. Qualitative comparison of the proposed method with single-image object detection network. (a)–(h) indicate the stenosis detection results of RetinaNet, Faster RCNN, Cascade R-CNN, FoveaBox, DETR, Deformable DETR, DINO and STQD-Det, respectively.

and they are less capable of capturing small targets. Therefore the Faster RCNN method produces a large number of missed detections on lesions ①, ③, and ⑤, while the Cascade R-CNN method mitigates this problem to a certain extent by cascading detectors and reduces the missed detections. Fig. 6(e) shows the DETR method, which cannot converge effectively within this dataset, and the prediction results are all initially generated detection query. Whereas the Deformable DETR method shown in Fig. 6(f) reduces the training cost of the network through the Deformable Attention Mechanism and can converge effectively. However, the lack of local induction bias in the Transformer network itself leads to its poor detection performance on some small-scale lesions. Fig. 6(g) shows the DINO method, which introduces Anchor into the Query based method to improve the detection performance. However, all the above methods lack the ability to utilize video sequence information, resulting in poor consistency of detection results within the same sequence. It also proves that the method proposed in this paper can fully exploit the temporal information contained in the video sequence images and improve the performance of the model in the video object detection task.

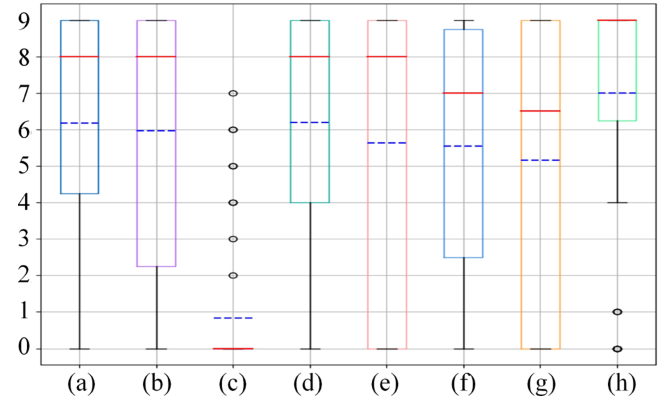


Fig. 7. STQD-Det and the single-image object detection methods can completely predict the correct frame boxplot in the same sequence. (a)–(h) indicate the results of Cascade R-CNN, Deformable DETR, DETR, DINO, Faster R-CNN, FoveaBox, RetinaNet and STQD-Det, respectively.

TABLE III
COMPARISON OF VIDEO OBJECT DETECTION METHODS

Methods	<i>Prec</i>	<i>Sens</i>	F_1	<i>FPS</i>
FGFA	89.76	73.96	81.10	6.91
DFF	86.36	61.30	71.63	23.02
SELSA	86.11	82.61	84.23	2.81
Temporal RoI Align	85.75	79.74	82.52	1.96
TransVOD	84.00	84.62	84.30	10.65
STQD-Det	94.52	90.46	92.45	25.08

To verify the consistency of the detection results between STQD-Det and single-image object detection methods in a same sequence, we tallied the instances where all methods made entirely accurate predictions in the same sequence. These results were subsequently condensed into a box plot, as depicted in Fig. 7. The blue dotted line represents the average of the number of completely correct frames, the red solid line represents the median, and the black circle represents the outlier. As depicted in the figure, the proposed method exhibits a median value of 9 and an average of 7. These statistics suggest that the proposed method significantly outperforms other single-frame target detection methods in terms of prediction accuracy, consistency, and stability. These results serve as evidence that the QD and STFS modules have a positive impact on enhancing the prediction stability of the network in the task of XRA sequence stenosis detection. The boxplot results of other methods are also consistent with the quantitative experimental results, which proves that the stability of sequence prediction is helpful to improve the accuracy of stenosis detection.

F. Comparison With Video Object Detection Network

In order to compare the ability to utilize the temporal information provided by XRA sequences of the proposed method and other video object detection (VOD) methods, which contains FGFA [42], DFF [43], SELSA [44], Temporal RoI Align [45] and TransVOD [46]. The results of quantitative comparison are shown in Table III. The FGFA and DFF both are optical flow-based algorithms, which will accumulate estimation errors

TABLE IV
COMPARISON OF STENOSIS DETECTION METHODS

Methods	<i>Prec</i>	<i>Sens</i>	F_1	<i>FPS</i>
SSD	22.89	79.88	35.57	0.01
H-CQN	40.67	77.03	53.20	1.14
DSSD	78.30	76.39	77.33	8.72
Stenosis-DetNet	87.01	87.91	87.45	15.51
STFA	92.27	89.56	90.88	2.53
STQD-Det	94.52	90.46	92.45	25.08

during the propagation process, resulting in poor *Sens*. Notably, DFF incorporates features from previous frames to enhance the current frame, which results in diminished accuracy when the image quality of the previous frame is suboptimal. This is evident from the fact that DFF's *Sens* is 29.16% lower than that of the proposed method. SELSA and Temporal RoI Align use temporal information aggregation instead of optical flow to avoid cumulative error and improve *Sens* significantly. However, these two methods are CNN-based and lack robust spatial long-distance context modeling capabilities. TransVOD uses transformer to obtain spatio-temporal context information, but the lack of local induction bias leads to the poor performance. In this paper, the STFS module obtains spatio-temporal features and performs best in *Prec*, *Sens* and F_1 . Furthermore, the proposed method exhibits superior inference speed. This advantage is attributed to the fact that SQNB are generated randomly by the system and directly input into the stenosis decoder for coordinate and class prediction. In contrast, other methods require anchor generation through RPN or employ a transformer to adjust learnable queries before feeding into the decoder. This additional complexity significantly hampers their time efficiency.

Qualitative results comparing the proposed method with five VOD approaches are presented in Fig. 8. FGFA and DFF exhibit certain issues illustrated in Fig. 8(a) and (b). Notably, these methods experience a loss of detection boxes in later frames due to feature degradation during the feature transfer process. In the case of DFF, the previous features are used to enhance the current frame, leads to FP detections in the last frame. The stenosis ① is occluded by other vessel in some frames. If don't enhance the feature, it will be difficult to detect. Taking TransVOD as an example, the transformer lacks local inductive bias, resulting in ignoring the weak stenosis lesion, resulting in FN detections. Stenosis ① also extends over a considerable length. However, the SELSA method and the Temporal RoI Align method, shown in images Fig. 8(c) and (d), rely solely on a CNN for feature extraction and enhancement. The limited receptive field of these methods hinders their ability to effectively detect long lesions, resulting in unstable detections. The proposed method enhances the characteristics of weak stenosis by STFS, thus achieving accurate detection.

G. Comparison With Stenosis Detection Network

To evaluate the performance of the proposed method in stenosis detection, we conducted a comparative analysis against five existing stenosis detection frameworks: Segmentation-based

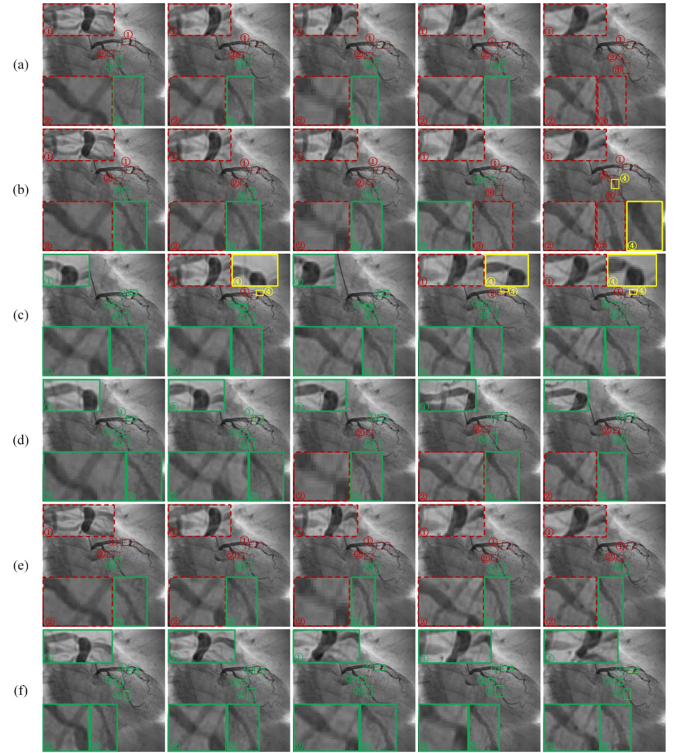


Fig. 8. Qualitative comparison between STQD-Det and the VOD methods. (a)–(f) denote the results of FGFA, DFF, SELSA, Temporal RoI Align, TransVOD and STQD-Det, respectively.

Stenosis Detection (SSD) [5], H-CQN [9], DSSD [8], Stenosis-DetNet [10] and STFA [11]. The results of quantitative comparison are shown in Table IV. SSD exhibits the poorest results across all metrics primarily due to its sensitivity to the accuracy of vessel segmentation. A low *Prec* indicates a high number of FP detections. This algorithm entails the extraction of coronary skeleton information from the segmentation result, which involves time-consuming data processing steps, making it the slowest method in terms of inference speed. H-CQN also produces a lot of FP detections, because this method uses quantum network to divide the input image into several sub-blocks, and then distributed training through quantum circuit. However, the randomly cut sub-blocks can only obtain local features of vessels without long-distance context information, resulting in low detection accuracy. DSSD and Stenosis-DetNet use the seq-fps module and the sequence consistency alignment module to aggregate the spatio-temporal context features, respectively. However, these two modules struggle to correlate temporal context information effectively for XRA images with substantial motion amplitude, leading to diminished detection accuracy. STFA uses transformer to aggregate spatio-temporal context information from RoI features, to improve the detection accuracy. Nevertheless, this method fails to boost features in wrongly detected frames, resulting fail detections in some frames. Additionally, due to the computational complexity of transformer, the inference speed of this method is slow. The

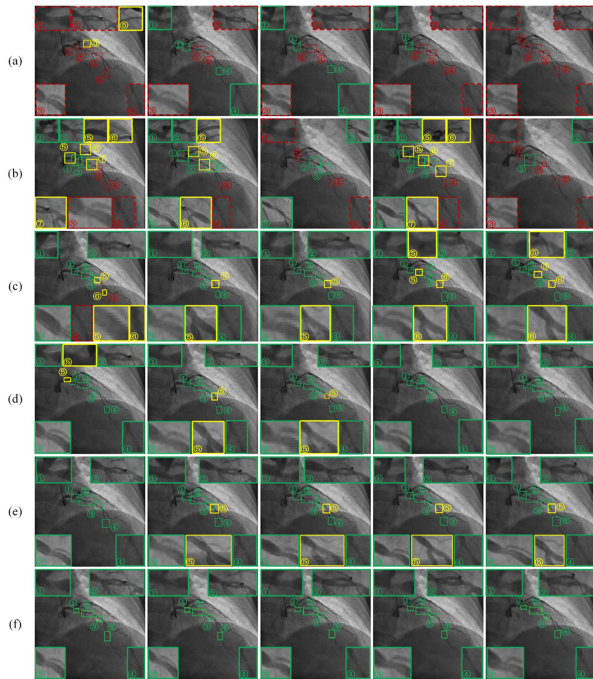


Fig. 9. Qualitative comparison between STQD-Det and the stenosis detection networks. (a)–(f) denote the results of SSD, H-CQN, DSSD, Stenosis-DetNet, STFA and STQD-Det, respectively.

proposed method not only aggregates the spatio-temporal context information through the STFA module. This enhancement significantly improves the accuracy of stenosis detection.

Fig. 9 shows the qualitative comparative experimental results of the method proposed in this paper and five coronary XRA stenosis detection methods. In this case, the challenge lies in the presence of multiple stenoses with varying lesion scales, making accurate stenosis detection more complex. For sequence (a), there are almost no correct detections. This outcome results from the method's reliance on vessel segmentation results. In segments where lesions ①, ②, and ③ are situated, the region with a normal radius is too small for the algorithm to calculate the radius accurately, leading to missed detections. In sequence (b), corresponding to the H-CQN method, the image is divided into several sub-blocks for stenosis detection, causing the network to lose access to long-distance context information. For a longer lesion such as lesion ②, it is split into two lesions during detection, resulting in FP detection. The results of DSSD and Stenosis-DetNet methods are shown in 9(c) and 9(d). Both methods cause more than half of the key frames with FP detections, because these two methods suppress FP detection in a non-learning way. When the vascular motion amplitude is large, lead that FP detection frame overlaps with the true positive detection frame in other frames, so that the algorithm cannot suppress the FP detection. Sequence (e) is the qualitative result of the STFA method. The detection results of this method for different frames are consistent, but there is a FP detection ⑤. This position is a bifurcation structure, which is divided into two thin branches by a thicker vessel. Without a robust understanding of stenosis, the network may produce false detections in such cases. The proposed method enhances the network's feature extraction and comprehension capabilities regarding stenosis

through STFS, effectively mitigating the occurrence of false detections.

V. CONCLUSION

In this paper, we propose a novel STQD-Det for accurate stenosis detection in coronary XRA sequences, utilizing a quantum noise diffusion model with spatio-temporal feature sharing. To enhance inference speed, we introduce a diffusion model based on quantum noise, generating discrete SQNB with prior sequence knowledge during the forward diffusion process. Addressing issues arising from poor image quality, vessel motion, and complex vascular structures, we introduce the STFS module that shares features from correct images into falsely detected frames. To validate the effectiveness of our approach, we conducted experiments using 233 coronary artery XRA images. Ablation experiments demonstrate the significant contributions of the SQNB and STFS modules in leveraging temporal information from XRA images and enhancing spatiotemporal features in erroneously detected images, ultimately improving stenosis detection performance in XRA images. Furthermore, we compared our proposed method with single-frame object detection networks, video object detection networks, and state-of-the-art methods for coronary artery stenosis detection. Our experimental results demonstrate that our method outperforms others in terms of detection accuracy and speed. In the clinical field, stenosis detection is only one aspect, and there are still challenges in the task of stenosis quantification. Therefore, in future work, we will extend our work to stenosis quantification. This will be achieved by incorporating stenosis grading heads into the proposed framework to facilitate real-time classification of stenosis severity, or by integrating the proposed framework with instance segmentation techniques to enable precise calculation of stenosis rates. Such advancements are essential to fully underpin clinical decision taking.

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